

Milind Verma

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EDUCATION

Vellore Institute of Technology, Bhopal

B.Tech in Computer Science Engineering – CGPA: 8.42/10.0

Bhopal, India

2023 – 2027

CBSE Senior Secondary School (12th)

Percentage: 82%

India

2023

CBSE Secondary School (10th)

Percentage: 88.40%

India

2021

TECHNICAL SKILLS

- **Programming Languages:** Java, Python, C++, JavaScript, PHP, HTML, CSS
- **Frameworks & Libraries:** React.js, Node.js, Scikit-Learn, Pandas, NumPy, Matplotlib, Seaborn, TensorFlow, Keras, OpenCV, Flask
- **Tools & Platforms:** AWS, Streamlit, Jupyter Notebook, Git, GitHub, VS Code, Docker, Postman, Cisco Packet Tracer
- **Databases:** MySQL, MongoDB, PostgreSQL, SQLite, Firebase
- **Core Concepts:** Data Structures & Algorithms, OOP, DBMS, Computer Networks, Operating Systems, Machine Learning

PROJECTS

Analyzing Urban Growth for Green Space Planning Using GIS

2026

Python, GIS, Deep Learning, U-Net, OpenCV, Flask

- Developed a GIS-based urban growth analysis system using semantic segmentation on satellite imagery for green space planning
- Trained a U-Net deep learning model for land-use classification and urban area segmentation using custom tiled datasets
- Implemented preprocessing, patch generation, prediction pipelines, and visualization modules for spatial analysis and growth prediction
- Built environmental recommendation and growth prediction modules to assist sustainable urban planning decisions
- Integrated prediction outputs, training curves, and GIS visualizations into an interactive Python-based application

Facial Emotion & Mental Health Detection

2025

Python, TensorFlow, Keras, OpenCV, Streamlit

- **Role:** Project Lead & Core Developer – Initiated after observing rising mental health concerns among students during college workshops, aiming to build an accessible early-detection tool
- Designed and implemented the complete pipeline: CNN-based deep learning model on the FER-2013 dataset classifying 7 emotions with 82% accuracy
- Developed real-time face detection using OpenCV Haar Cascades integrated with the trained model to classify emotions from live webcam feed
- Created custom functions `preprocess_image()`, `predict_emotion()`, and `map_to_wellbeing()` for emotion analysis and mental health correlation
- Built an interactive Streamlit interface for image upload and live webcam analysis, with visual probability graphs and well-being recommendations
- Handled end-to-end workflow: data preprocessing, augmentation, model training, hyperparameter tuning, evaluation, and deployment

SuMoPy (Sustainable Mobility with Python)

2024

Python, Data Analysis, NetworkX, Matplotlib, Optimization Algorithms

- **Role:** Team Member & Algorithm Developer – Contributed to optimization and visualization modules under a faculty-led green technology initiative
- Developed a route optimization system promoting eco-friendly urban transportation using travel datasets
- Implemented Dijkstra's and A* algorithms inside `find_shortest_route()` and `calculate_emissions()` considering distance, time, and CO₂ emissions
- Built Matplotlib dashboards for travel patterns, route comparisons, and environmental impact analysis
- Collaborated with a 3-member team handling algorithm design, testing, and integration

CERTIFICATIONS

- **The Bits and Bytes of Computer Networking** – Google/Coursera (2024)
- **Introduction to Machine Learning** – NPTEL (2024)
- **Networking Basics** – Cisco Networking Academy (2025)
- **Networking Devices and Initial Configuration** – Cisco Networking Academy (2025)

EXTRA-CURRICULAR ACTIVITIES

President, Null Student Chapter

Jun 2024 – Present

VIT Bhopal

- Led technical discussions and coordinated cybersecurity workshops for 100+ students, fostering a culture of security awareness
- Organized chapter events, managed team responsibilities, and collaborated with industry professionals for guest sessions